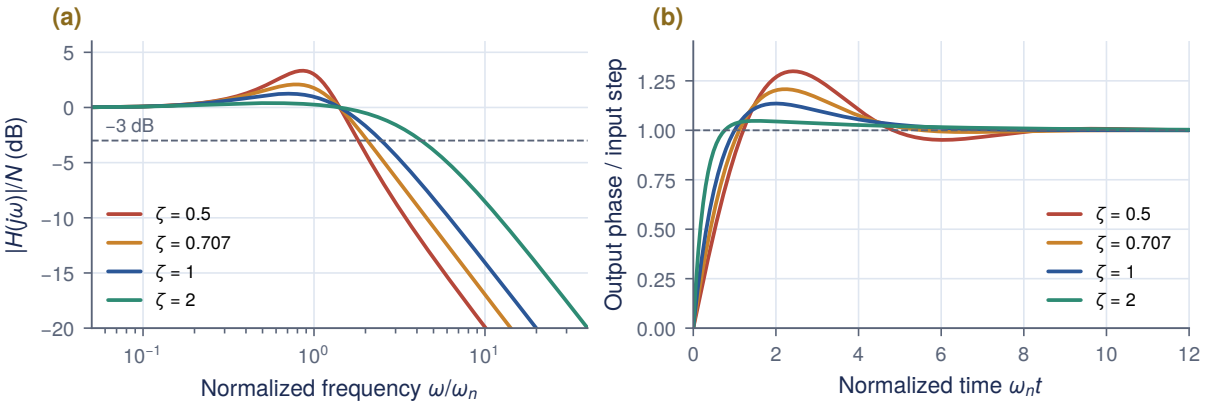
(a)
5
0
−5
−10
−15
−20
|H(jω)|/N (dB)
−3 dB
ζ = 0.5
ζ = 0.707
ζ = 1
ζ = 2
10^−1
10^0
10^1
Normalized frequency ω/ω_n
(b)
1.25
1.00
0.75
0.50
0.25
0.00
Output phase / input step
ζ = 0.5
ζ = 0.707
ζ = 1
ζ = 2
0
2
4
6
8
10
12
Normalized time ω_n t